

Bryan C. Busby

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Education

B.Sc. in Pure Mathematics, University of Puerto Rico 2023–2026

Research Experience

Villanova University (REU) – Advisor: Alexander Díaz-López Summer 2025

Worked with other undergraduates on arithmetical structures on graphs, especially cycles with multiple edges.

UPR – Advisor: Omar Colón-Reyes Spring 2024

Studied transient-length bounds for Boolean monomial systems by graph-theoretic methods, including walks and the Frobenius number.

UPR (CARSE) – Advisor: Rafael Rodríguez Solís 2023

Analyzed the effects of instrument tolerances on 5G propagation measurements. Debugged software–hardware communication in GNU Radio.

High School – Advisor: Edwin Benvenutti 2021

Studied powers of adjacency matrices and their relation to walks on graphs.

Manuscripts in Preparation

Arithmetical Structures on Cycles with a multi-edge (with Alexander Díaz-López)

Transient Length Boundary for Finite Dynamical Systems (with Omar Colón-Reyes)

Presentations

Arithmetical Structures on Cycles with a multi-edge – MAA MathFest, Sacramento, CA Aug 2025

Finite Dynamical Systems: Transient Bounds for Boolean Monomial Systems – SIDIM May 2024
(PR Interdisciplinary Mathematics Symposium), UPR Ponce

Development of Radio Wave Propagation Setup and Propagation Measurements at 5G Bands 2023
– UPR Mayagüez

On the Adjacency Matrix and Walks – Regional Math Fair (PR Dept. of Education) 2021

Teaching Experience

Private Mathematics Tutor 2022–present

Tutored students individually in high-school and early undergraduate mathematics. Completed more than 100 hours of instruction.

Employment

Production Worker 2026–present

Food Server 2025–2026

Food Server 2023–2024

Languages

Spanish (native); English (fluent).